

Revision Notes for Class 8 Science Chapter 6 – Winds, Storms and Cyclones

Understanding pressure, winds, storms, and cyclones is essential for grasping the forces shaping weather and climate. In daily life, we observe objects being moved by wind, water tanks on rooftops, and the impact of storms, but the science behind these phenomena is deeply rooted in the concepts of pressure and movement of air.

Pressure and Its Effects

Pressure is the amount of force exerted per unit area. If a force is applied over a small area, the pressure is high, and if the same force is spread over a larger area, the pressure decreases. This basic idea explains why a bag with broad straps feels more comfortable than one with narrow straps and why buckets with broad handles are easier to carry than those with thin handles. The SI unit of pressure is the pascal (Pa), equal to one newton per square metre (N/m^2).

In many daily activities, pressure determines ease or difficulty. A sharp knife cuts fruit more easily than a blunt one because the sharp edge applies force over a smaller area, increasing pressure. Similarly, an iron nail with a pointed tip enters wood quickly, while its flat head makes hammering difficult.

Pressure in Liquids

Liquids, like water, also exert pressure. The pressure at a point in a liquid depends on the height of the liquid column above it, not just on the total volume. For example, when two glass tubes of different widths are filled to the same height with water and have balloons attached at the bottom, the balloons bulge equally. This shows that greater height, not width, means more pressure.

If you increase the water level in a tank placed on a roof, the pressure in the taps below increases, providing a stronger stream of water. This is why water tanks are placed at

heights in buildings. The greater the height, the greater the pressure at the bottom floors.

Liquids also exert pressure sideways, not just downwards. If holes are made at the same height on the sides of a bottle filled with water, water will flow out through all the holes, proving that pressure in liquids acts in all directions. This is also the reason why the base of a dam is made broader than the top—to withstand the large water pressure at the bottom.

Air Pressure and Atmosphere

Air has weight and exerts pressure on everything in contact with it. The pressure exerted by the air around us is called atmospheric pressure. It can be quite large; a 15 cm x 15 cm area experiences a force equal to the weight of roughly 225 kg. However, we do not feel crushed by this pressure because our bodies exert an equal internal pressure outward, balancing the effects.

Common activities like trying to pull off a rubber sucker stuck to a smooth wall, or noticing how a blown-up balloon pushes outwards in all directions, are examples of air pressure in action. Air pressure decreases with height, which impacts the weather and our daily experiences.

How Winds are Formed

Winds are caused by differences in air pressure. Air always moves from regions of high pressure to low pressure. For example, during the daytime near the sea, land heats up faster than water, creating a low pressure zone. Cooler air from the sea moves inland to replace the rising warm air—this movement is called the sea breeze. At night, the opposite happens because land cools down faster than sea, and the breeze moves from land to sea (land breeze).

Air pressure is measured in units such as millibar (mb), pascal (Pa), and hectopascal (hPa), with 1 mb = 100 Pa. These differences drive local and global wind patterns.

Effect of High-Speed Winds

When wind moves very quickly, it creates a low-pressure area. If you hang two inflated balloons a few centimeters apart and blow air between them, the balloons move closer together. This occurs because the air you blow creates low pressure between the balloons, and higher pressure from outside pushes them together. Similarly, high-speed winds can lift roofs from houses by creating low pressure above, while higher pressure inside pushes the roof upward. Keeping doors and windows open during storms allows pressure inside and outside to even out, reducing damage.

Thunderstorms and Lightning

Thunderstorms form when warm, moist air rises quickly, cools, and condenses to form clouds. The rapid upward and downward movement of air separates positive and negative charges in clouds. When the charge difference becomes large enough, a sudden electric discharge—lightning—occurs, followed by a thunder sound as the air expands rapidly. Safety during thunderstorms includes avoiding open fields, metallic objects, and water bodies. Staying in vehicles or indoors is generally safer.

Lightning conductors, often fitted to tall buildings, help protect structures by safely transferring the electric charge from lightning into the ground.

Cyclones

Cyclones are very powerful storms formed over warm ocean waters. Warm, moist air rises over the ocean, creating a centre of very low pressure known as the eye of the cyclone. Air from surrounding areas rushes in to fill the space, and the rotation of the earth makes the cyclone spin. When cyclones make landfall, they bring destructive winds, heavy rainfall, and flooding, resulting in significant damage to lives and property.

Some examples, like cyclone Amphan, reached wind speeds of 270 km/h. Preparedness includes following local weather alerts, having emergency kits, and moving to safe cyclone shelters if instructed.

Key Learnings and Important Points

- Pressure is the force acting per unit area. More area means less pressure with the same force.
- The SI unit of pressure is the pascal ($1 \text{ Pa} = 1 \text{ N/m}^2$).
- Liquids and gases apply pressure in all directions on the walls and bottom of containers.
- Atmospheric pressure is the air pressure exerted by the layers of air on everything at Earth's surface.
- Wind is air moving from high pressure to low pressure regions.
- High-speed winds lower air pressure, causing various effects, such as roofs getting blown away or balloons moving together.
- Thunderstorms need moist air, strong upward winds, and often cause lightning, which can be dangerous.
- Cyclones are intense storms with low atmospheric pressure at the centre, leading to very strong winds.
- Staying safe involves following warnings, moving to shelters, and keeping away from tall trees and power lines in storms.

Practice and Application

- If an iron nail is hammered into wood, it goes in easily when the pointed edge is used because of higher pressure on a smaller surface.
- If you are filling water balloons attached to vertical pipes of the same height but different thickness, both balloons bulge equally—height controls pressure, not the width.
- When two boats carry different numbers of people but have different base areas, the one with less area and more total weight experiences more pressure and is at greater risk of sinking.